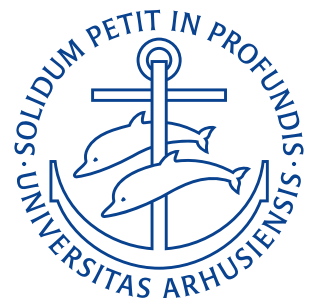


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Lecture Notes

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Lecture 1: Introduction

17. Marts 2026

Example: Partial Fraction Decomposition

$$\begin{aligned}
 G(s) &= \frac{5s+3}{(s+1)(s+2)(s+3)} \\
 \frac{5s+3}{(s+1)(s+2)(s+3)} &= \frac{K_{s1}}{s+1} + \frac{K_{s2}}{s+2} + \frac{K_{s3}}{s+3} \\
 K_{-1} &= \frac{p(-1)}{(-1+2)(-1+3)} = -1 \\
 K_{-2} &= \frac{p(-2)}{(-2+1)(-2+3)} = 7 \\
 K_{-3} &= \frac{p(-3)}{(-3+1)(-3+2)} = -6 \\
 \frac{5s+3}{(s+1)(s+2)(s+3)} &= -\frac{1}{s+1} + \frac{7}{s+2} - \frac{6}{s+3} \\
 G(t) &= -e^{-t} + 7e^{-2t} - 6e^{-3t}.
 \end{aligned}$$

Example: Conversion of ODE to Algebraic Equation using Laplace Transforms

Given the ODE

$$\frac{d^2 y}{dt^2} + 6 \frac{dy}{dt} + 8y = 2 \quad y(0) = y'(0) = 0$$

we want to find a solution using the Laplace transform to turn the above ODE into an easily-solvable algebraic equation.

We start by finding the Laplace transform of each term.

$$\begin{aligned}
 \frac{d^2 y}{dt^2} &= s^2 Y(s) - sy'(0) - y'(0) \\
 6 \frac{dy}{dt} &= 6(sY(s) - y(0)) \\
 8y &= 8Y(s) \\
 2 &= \frac{2}{s} \\
 s^2 Y(s) - 0 - 0 + 6(sY(s) - 0) + 8Y(s) &= \frac{2}{s} \\
 s^2 Y(s) + 6sY(s) + 8Y(s) &= \frac{2}{s} \\
 Y(s)(s^2 + 6s + 8) &= \frac{2}{s} \\
 Y(s) &= \frac{2}{s(s+2)(s+4)}.
 \end{aligned}$$

We have now found the solution in the Laplace-domain. To find the corresponding time-domain solution

we take the inverse Laplace transform. To do this we use Partial Fraction Decomposition:

$$\begin{aligned}\frac{2}{s(s+2)(s+4)} &= \frac{K_{s1}}{s} + \frac{K_{s2}}{s+2} + \frac{K_{s3}}{s+4} \\ K_0 &= \frac{2}{2 \cdot 4} = \frac{1}{4} \\ K_{-2} &= \frac{2}{-2 \cdot 2} = -\frac{1}{2} \\ K_{-4} &= \frac{2}{-4 \cdot -2} = \frac{1}{4} \\ \frac{2}{s(s+2)(s+4)} &= \frac{1}{4s} - \frac{1}{2(s+2)} + \frac{1}{4(s+4)}\end{aligned}$$

Now, that a partial fraction decomposition has been made we can easily take the inverse laplace of the solution as

$$\begin{aligned}g(t) &= \frac{1}{4} - \frac{1}{2}e^{-2t} + \frac{1}{4}e^{-4t} \\ &= \frac{1}{4}e^{-4t}(e^{2t} - 1)^2.\end{aligned}$$

Example: In-Class Exercise: Controlling a Laser Printer

A laser printer uses a laser beam to print copy rapidly for a computer. The laser is positioned by a control input $r(t)$, so that we have

$$Y(s) = \frac{4(s+50)}{s^2 + 30s + 200} R(s).$$

The input $r(t)$ represents the desired position of the laser beam.

- If $r(t)$ is a unit step input find the output $y(t)$.
- What is the final value of $y(t)$?

(a) For a unit step $r(t)$, $R(s) = \frac{1}{s}$, so

$$\begin{aligned}Y(s) &= \frac{4(s+50)}{s^2 + 30s + 200} \frac{1}{s} \\ &= \frac{4(s+50)}{s^3 + 30s^2 + 200s} \\ &= \frac{4(s+50)}{s(s+10)(s+20)} \\ \frac{4(s+50)}{s(s+10)(s+20)} &= \frac{K_0}{s} + \frac{K_{10}}{s+10} + \frac{K_{20}}{s+20} \\ K_0 &= \frac{200}{10 \cdot 20} = 1 \\ K_{-10} &= \frac{160}{-10 \cdot 10} = -\frac{8}{5} \\ K_{-20} &= \frac{120}{-20 \cdot -10} = \frac{3}{5} \\ Y(s) &= \frac{1}{s} - \frac{8}{5(s+10)} + \frac{3}{5(s+20)} \\ y(t) &= 1 - \frac{8}{5}e^{-10t} + \frac{3}{5}e^{-20t}.\end{aligned}$$

(b) We solve:

$$\lim_{t \rightarrow \infty} 1 - \frac{8}{5}e^{-10t} + \frac{3}{5}e^{-20t} = 1.$$

Lecture 2: Mathematical Models of Systems

20. Marts 2026

2.1 Block Diagrams

Though mathematical models are useful in describing the complex behavior of a control system they are often rather unintuitive. A more graphical alternative to the rigorous mathematical models are Block Diagrams. An example of one of these is shown on [Figure 2.1](#)

Definition 1: Block Diagram

A *Block Diagram* is a schematic representation of the cause-and-effect relationship throughout the system together with the transfer functions of the variables of interest.

Typical elements of block diagrams include but are not limited to:

- Comparators
- Blocks representing individual component transfer functions (sensors, actuators, controllers, plants, etc.)
- Input or reference signals
- Output signals
- Disturbance signals
- Feedback loops

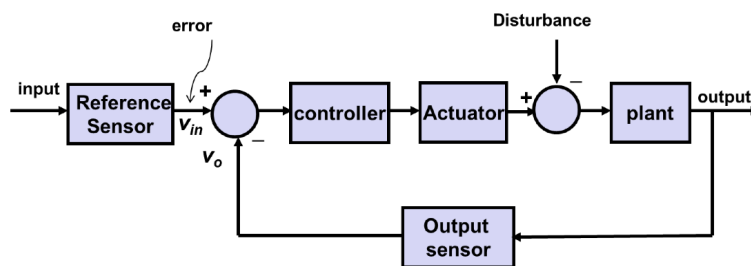


Figure 2.1: Example of a block diagram.

An example showing how different components of a system might be represented as blocks of a block diagram is shown in [Figure 2.2](#).

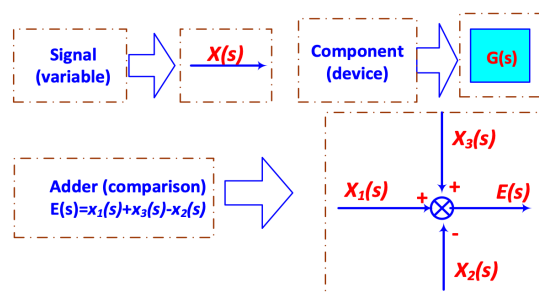


Figure 2.2: Different components represented as blocks.

For multiple input or output systems the subscript convention is such that G_{ij} is the transfer function of output i and input j .

2.1.1 Transformation and Reduction of Block Diagrams

Different Block diagrams can be transformed and reduced into more simple forms. This is especially useful for large and complex systems.

For a simple cascade block diagram as shown on Figure 2.3 one is able to reduce the block diagram into a single $G_1 \cdot G_2$ -block.



Figure 2.3: Cascade Block diagram.

For a parallel block diagram as shown on Figure 2.4 the resulting transformation is a single $G_1 + G_2$ -block.

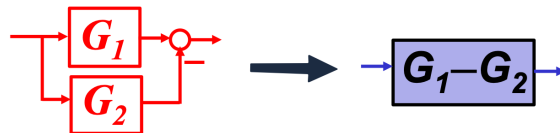


Figure 2.4: Parallel block diagram.

For a negative feedback Block diagram as shown on Figure 2.5 the reduction is a single block:

$$\frac{G_1}{1 + G_1 G_2}.$$

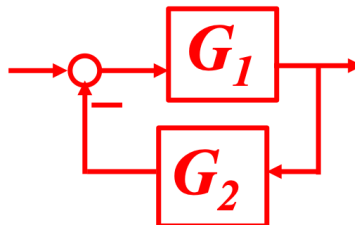


Figure 2.5: Feedback block diagram.

A summing node is able to be moved across a block as shown in Figure 2.6.

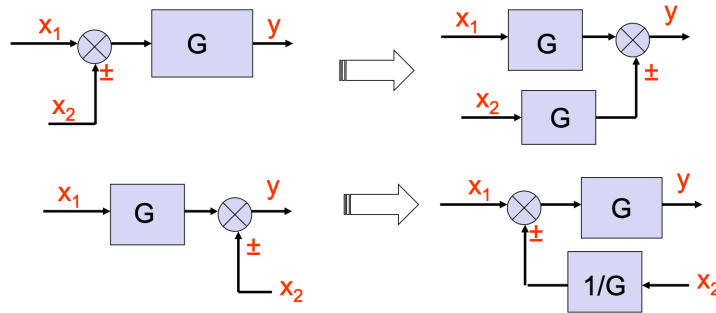


Figure 2.6: Movement of a summing node across a block.

Pickoff points can also be moved across blocks. The manner this is done by is shown in Figure 2.7.

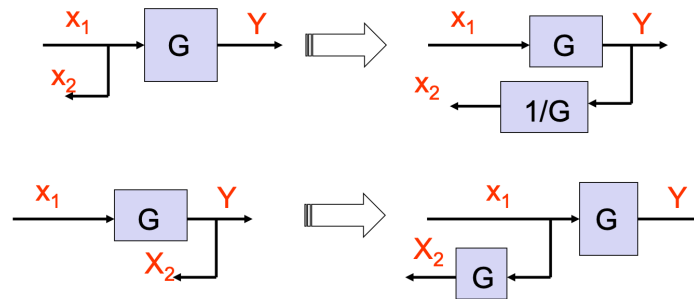


Figure 2.7: Movement of a pickoff point across a block.

It is also possible to interchange neighboring summing points or pickoff points as shown in Figure 2.8.

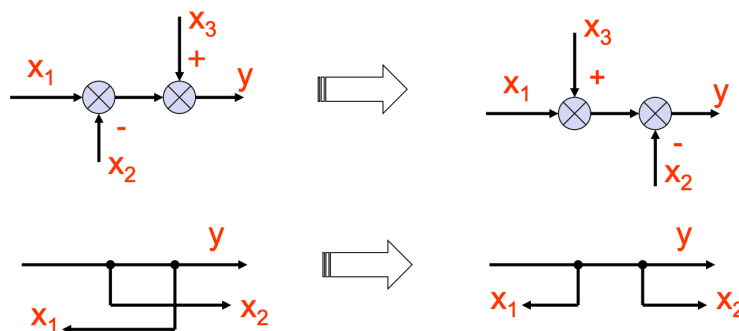


Figure 2.8: Interchanging neighboring summing and pickoff points

Example: Transfer function of an interacting system

Given the system shown find the transfer function.

We combine all the feedback loops using:

$$G_i(s) \parallel H_i(s) = \frac{G_i(s)}{1 - G_i(s)H_i(s)}.$$

This gives:

$$\begin{aligned}
 G_2(s) \parallel H_2(s) &= \frac{G_2(s)}{1 - G_2(s)H_2(s)} \\
 G_3(s) \parallel H_3(s) &= \frac{G_3(s)}{1 - G_3(s)H_3(s)} \\
 G_6(s) \parallel H_6(s) &= \frac{G_6(s)}{1 - G_6(s)H_6(s)} \\
 G_7(s) \parallel H_7(s) &= \frac{G_7(s)}{1 - G_7(s)H_7(s)}.
 \end{aligned}$$

Now the two branches are both cascading, so using $G_1 G_2$ for series we get:

$$\begin{aligned}
 B_1(s) &= G_1(s) \cdot \frac{G_2(s)}{1 - G_2(s)H_2(s)} \cdot \frac{G_3(s)}{1 - G_3(s)H_3(s)} \cdot G_4(s) \\
 B_2(s) &= G_5(s) \cdot \frac{G_6(s)}{1 - G_6(s)H_6(s)} \cdot \frac{G_7(s)}{1 - G_7(s)H_7(s)} \cdot G_8(s).
 \end{aligned}$$

Now, we are left with the two parallel $B_1(s)$ and $B_2(s)$ branches which are combined using $G_1 \parallel G_2 = G_1 + G_2$ as:

$$B_1(s) + B_2(s) = G_1(s) \cdot \frac{G_2(s)}{1 - G_2(s)H_2(s)} \cdot \frac{G_3(s)}{1 - G_3(s)H_3(s)} \cdot G_4(s) + G_5(s) \cdot \frac{G_6(s)}{1 - G_6(s)H_6(s)} \cdot \frac{G_7(s)}{1 - G_7(s)H_7(s)} \cdot G_8(s).$$

Our transfer function is therefore:

$$t(s) = G_1(s) \cdot \frac{G_2(s)}{1 - G_2(s)H_2(s)} \cdot \frac{G_3(s)}{1 - G_3(s)H_3(s)} \cdot G_4(s) + G_5(s) \cdot \frac{G_6(s)}{1 - G_6(s)H_6(s)} \cdot \frac{G_7(s)}{1 - G_7(s)H_7(s)} \cdot G_8(s).$$

So

$$Y(s) = R(s) \left(G_1(s) \cdot \frac{G_2(s)}{1 - G_2(s)H_2(s)} \cdot \frac{G_3(s)}{1 - G_3(s)H_3(s)} \cdot G_4(s) + G_5(s) \cdot \frac{G_6(s)}{1 - G_6(s)H_6(s)} \cdot \frac{G_7(s)}{1 - G_7(s)H_7(s)} \cdot G_8(s) \right)$$